

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

Semester - I

Broad Category of Course	Semester 1		Teaching Scheme (Hours/Week)				Credits/ Grades				Examination Scheme and Marks					
	Subject code	Name of subjects	L	P	T	Total	L	P	T	Total	Theory			Practical		Total
											ISE	CIE	ESE	CIE/ TW	ESE (P/OR)	
PCC	2661PC11	Mathematical Foundations for Computing	3	-	-	3	3	--	-	3	20	30	50	--	--	100
PCC	2671PC12	Advanced Machine Learning	3	-	-	3	3	--	--	3	20	30	50	--	--	100
PCC	2671PC13	Advanced Data Management and Warehousing	4	-	-	4	4	--	--	4	20	30	50	--	--	100
HSS	2661RM11	Research Methodology	3	-	-	3	3	--	--	3	20	30	50	--	--	100
PEC	2671PE1X	Program Elective - I	3	-	-	3	3	--	--	3	20	30	50	--	--	100
VSEC	2671VS31	Laboratory Proficiency - I	-	4	-	4	--	2	--	2	--	--	--	25	25	50
HSS	2671RM31	Research and Technical Communication Lab	-	2	-	2	--	1	--	1	--	--	--	25	--	25
VSEC	2671VS32	Programming Proficiency Lab	-	2	-	2	--	1	--	1	--	--	--	25	--	25
		Total	16	8	-	24	16	4	-	20	100	150	250	75	25	600

2671PE1X: Program Elective I	
2671PE11: IoT Architecture & Protocols	2671PE12: Information Security
2671PE13: Natural Language Processing	2671PE14: Distributed Operating System

2671VS31: Laboratory Proficiency – I
Laboratory Assignments will be based on Program Core and Program Elective Courses.

Semester I

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27)			
Data Science (DS)			
[2661PC11] : Mathematical Foundations for Computing			
Semester	Credits	Teaching Scheme	Examination Scheme
1	3	L: 3 Hrs./ Week	CIE: 30 Marks ISE: 20 Marks ESE: 50 Marks
Prerequisite: Students should have prior knowledge of Fundamentals of Mathematics			
Course Objectives: The objective of this course is to provide students with - Basics of discrete mathematics for computing in computer science/Data Science applications. - Various data structures and algorithmic solutions for storage and retrieval of data computational problems. - Knowledge of statistical measures and hypotheses for computing. - Probability usage for solving computer science/Data Science problems. - Basics of linear algebra and calculus for solving computational problems.			
Course Outcomes: After completing this course, students will be able to: C01: Apply discrete mathematics concepts like sets, relations to solve computing problems. C02: Choose suitable tree and graph data structures and its algorithms for solving problems in various domains. C03: Perform exploratory data analysis and statistical validation by calculating measures of central tendency, dispersion, and correlation, and applying appropriate hypothesis tests to ensure data readiness for further modeling. C04: Select and apply appropriate probability distribution functions (PDFs) and probabilistic models to solve clustering and classification problems, ensuring the model's assumptions align with the underlying data characteristics. C05: Apply matrix algebra, eigenvalue decomposition, and multivariate calculus to solve systems of linear equations and optimize functions, specifically within the context of dimensionality reduction.			
COURSE CONTENTS			
Module-I	Discrete Mathematics for Computing		12 Hrs.

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43**M. Tech. (Data Science) Curriculum Structure****With effect from the A.Y. 2026-27****Set Theory:** Elementary Theory of Sets, Set Rules & Sets Combinations.**Relations:** Binary Relation, Equivalence Relation, Composite Relation, Partial Ordering Relation.**Fundamentals of Graph Theory:** Types, graph representation methods, Traveling-salesman problem, applications.**Trees & Search Trees:** balanced search trees, Multiway balanced search trees, splay tree, Trie, KD Tree**Heaps:** Binomial Heaps, Fibonacci Heap.**Case Study:** Mathematical modeling for applications of Graph Electrical network problem/ Flow graph notation/ Test case generation using graphs/Trie for spell checker/autocomplete string/ KD Tree for geographic information systems (GIS).**Module-II****Statistical Inference****9 Hrs.**

Types of Statistical Inference, Descriptive Statistics, Inferential Statistics, Importance of Statistical Inference in computing. Measures of Central Tendency: Mean, Geometric mean, Harmonic mean, Median, Mode, Midrange, Measures of Dispersion: Range, inter quartile range, Variance, Mean Deviation, Standard Deviation. Coefficient of variation: Moments, Skewness, Kurtosis, Pearson Correlation. Measures of relationship: Covariance, Pearson Correlation. Hypothesis testing: **Parametric Tests:** Means, for Differences between Means, for Comparing Two Related Samples of Proportions, for Difference between Proportions, for Comparing a Variance to Some Hypothesized Population Variance. Chi-square Test, and Analysis of variance and covariance. **Nonparametric Tests:** Sign Test, Wilcoxon (Signed Rank).

Case study: Study datasets like Iris/Diamond/Churn and apply suitable central tendency, dispersion and relationship measures to preprocess the datasets. Use of open-source statistical tools for hypothesis testing.

Module-III**Probability Theory****9 Hrs.**

Elements of Probability Theory: Basic Concept, Conditional Probability. Random variables: Cumulative Distribution Function, Expectation and Variance of Random variables, Covariance & Correlation. Probability Distributions: Discrete Uniform Distribution, Bernoulli, Binomial, Poisson, Geometric, and Hyper geometric Distributions. Continuous Distribution: Continuous Uniform Distribution, Normal Distribution, Exponential Distribution. Inductive statistics: Maximum Likelihood Estimation. Bayes' Theorem, Joint Probability, Bayesian network. Markov model.

Case Study: Bayesian Network for Spatial data/Markov model for product lifecycle

Module-IV**Linear Algebra & Vector Calculus****9 Hrs.**

Matrix and vector algebra, systems of linear equations using matrices, linear independence, Matrix factorization concept/LU decomposition, Eigenvalues and eigenvectors. Understanding of calculus: concept of function and derivative, Multivariate calculus: concept, Partial Derivatives, chain rule.

Case Study: System of linear equations for Electrical Networks/Eigen values & Eigen vector for Pagerank and PCA algorithm/ Gradient Descent

Text Books:

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

T1. K.H. Rosen, <i>Discrete Mathematics and its application</i> , Tata McGraw Hill, 2019, 8 th edition, ISBN 978-1-259-67651-2.
T2. Thomas H. Cormen, Charles E. Leiserson, Ronald L., Rivest Clifford Stein, <i>Introduction to Algorithms</i> , The MIT Press, 2022, 4 th Edition, ISBN 9780262046305.
T3. Richard A. Johnson, Irwin Miller, John Freund, <i>Probability and Statistics for Engineers</i> , Pearson Education, 2013, 9th Edition, ISBN 978-0-321-98624-5.
T4. Peter Brass, <i>Advanced Data Structures</i> , Cambridge University Press, 2008, 1st Edition, ISBN: 978-1-107-43982-5 / ISBN 978-0-511-43685-7.
Reference Books:
R1. Tremblay and Manohar, <i>Discrete Mathematical Structures with Applications to Computer Science</i> , Tata McGraw Hill, 1997, 1st Edition, ISBN-13: 978-0-07-463113-3.
R2. Irwin Miller, Marylees Miller, <i>Mathematical Statistics with Applications</i> , Pearson Education, 2014, 8th Edition, ISBN -9789332519053.
R3. Joe Mott, Abraham Kandel, <i>Discrete Mathematics and its Application</i> , Prentice Hall, 1995, 2 nd Edition, ISBN 978-81-203-1502-0.
R4. David C. Lay, Steven R. Lay, and Judi J. McDonald, <i>Linear Algebra and its Applications</i> , Pearson, 2015, Publication, 5th Edition, ISBN 978-0-321-98238-4.
Relevant MOOCs Course (Course name and Weblink)
1. Christopher Krapu, Robert Stewart, and Amy Rose. 2023, "A Review of Bayesian Networks for Spatial Data," in ACM Transaction on Spatial Algorithms System, Vol.9, Issue 1, Article 7 (January 2023), 21 pages. https://doi.org/10.1145/3516523 .
2. L. Wan, F. Xia, X. Kong, C. -H. Hsu, R. Huang and J. Ma, "Deep Matrix Factorization for Trust-Aware Recommendation in Social Networks," in IEEE Transactions on Network Science and Engineering, vol. 8, no. 1, pp. 511-528, 1 Jan.-March 2021, doi: 10.1109/TNSE.2020.3044035.
3. Emmert-Streib, Frank, and Matthias Dehmer, "Understanding Statistical Hypothesis Testing: The Logic of Statistical Inference," in MDPI Journal of Machine Learning and Knowledge Extraction, vol. 1, no. 3, pp. 945-961, doi: https://doi.org/10.3390/make1030054 .
Relevant Topics for Self-study:
Other Resources/Links
1. https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/



SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27)			
Data Science			
2671PC12 : Advanced Machine Learning			
Semester	Credits	Teaching Scheme	Examination Scheme
1	3	L: 3 Hrs./ Week	CIE: 30 Marks ISE: 20 Marks ESE: 50 Marks
Prerequisite: Students should have prior knowledge of Mathematics, Data Mining.			
Course Objectives: The objective of this course is to provide students with - Understand basic concepts and possible applications of machine learning. - Study and illustrate various data preprocessing methods and Supervised, Unsupervised machine learning techniques. - Study the basic principles of reinforcement learning - Understand the basic principles of optimization and its importance in machine learning.			
Course Outcomes: After completing this course, students will be able to: CO1: Describe the basic concepts and possible applications of machine learning. CO2: Evaluate classification, regression and clustering methods for real world applications for specified data. CO3: Apply reinforcement algorithms to real world problems. CO4: Solve Optimization problems in machine learning.			
COURSE CONTENTS			
Module-I	Introduction to Machine Learning and Regression		12 Hrs.
Models of Machine learning: Geometric model, Probabilistic Models, Logical Models, Grouping and grading models, Parametric and non-parametric models Features: Concept of Feature, Feature Extraction, Feature Selection Data Preprocessing and Dimensionality Reduction: Data integration, data reduction, concept hierarchy generation, PCA, LDA. Regression: Introduction, Multivariate Linear regression, Logistic regression, Lasso and Ridge Regression. Assessing performance of regression – Error measures, Overfitting.			

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

Module-II	Classification and Clustering algorithms	9 Hrs.
Classification: Classification concepts, Binary and multi-class classification, K nearest neighbor, Decision Tree Representation, Alternative measures for selecting attributes, ID3 algorithm, Naïve Bayes, Support Vector Machines as a linear and non-linear classifier. Ensemble methods: Introduction, Bagging, Boosting and Adaboost, Random Forest. Clustering: Introduction, Distance based clustering- K-means algorithm, Hierarchical Clustering-Agglomerative and Hierarchical, Expectation Maximization methods, Choosing number of clusters- silhouettes. Performance measure metric: Mean Square Error, R-squared, Precision, Recall, F1-score, Accuracy.		
Module-III	Reinforcement learning	9 Hrs.
Elements of Reinforcement Learning, Model-Based Learning, Dynamic programming, Monte Carlo, Temporal Difference Learning, Generalization. Bandit algorithms: Upper Confidence Bound algorithm. Case Study: Incremental Document Classification.		
Module-IV	Optimization Algorithms	9 Hrs.
Convex Functions, First and Second Order Conditions for Optimisations, Convex and Non-Convex Optimisation, problems in Machine Learning. Gradient Descent: math, programming basic optimisation problems and their solutions, Variants of Gradient Descent: Projected, Stochastic, Proximal, Accelerated, Coordinate Descent. Case Study: Stochastic Gradient Descent for Employee Attrition		
Text Books:		
T1. Ethem Alpaydin, <i>Introduction to Machine Learning</i> , PHI, 2013, 2nd edition, ISBN : 978-0-262-01243-0.		
T2. Kevin Murphy, <i>Machine Learning: A Probabilistic Approach</i> , MIT Press, 2012, 1st Edition, ISBN: 978-0262-30616-4.		
T3. C. M. Bishop, <i>Pattern Recognition and Machine learning</i> , Springer, 2013, 1st Edition, ISBN.: 978-81-322-0906-5.		
T4. Josh Patterson, Adam Gibson, <i>Deep Learning: A Practitioner's Approach</i> , O'Reilly Media, Inc, 2017, 1st Edition, ISBN No.: 9781491914250		
Reference Books:		
R1. Jiawei Han, Micheline Kamber, and Jian Pie, <i>Data Mining: Concepts and Techniques</i> , Elsevier Publishers, 2011, 3rd Edition, ISBN: 9780123814791, 9780123814807.		
R2. Hastie, Trevor, et al., — <i>The elements of statistical learning: Data mining, inference, and prediction</i> , Vol. 2. New York: Springer, 2009, 2nd Edition, ISBN- 978-0-387-84857-0.		
R3. Peter Flach, <i>Machine Learning: The Art and Science of Algorithms that make sense of data</i> , Cambridge University Press, 2012, 1st Edition, ISBN: 978-1-316-50611-0.		
R4. Hastie, Tibshirani, Friedman, <i>Introduction to statistical machine learning with applications in R</i> , Springer, 2013, 2nd Edition, ISBN: 978-1-4614-7138-7.		
R5. Tom Mitchell, <i>Machine Learning</i> , McGraw Hill, 1997, 1st Edition, ISBN: 0-07-042807-7.		

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

R6. Parag Kulkarni, "Reinforcement and Systemic Machine learning for Decision Making," Wiley-IEEE Press, 2012, 1st Edition, ISBN: 978-0-470-91999-6.
R7. S. Boyd and L. Vandenberghe, <i>Convex optimization</i> , Cambridge university press, 2004, 1st Edition, ISBN: 978-0521833783 .
Relevant MOOCs Course (Course name and Weblink)
1. Introduction to Machine Learning by Sudeshna Sarkar IIT Kharagpur https://nptel.ac.in/courses/106105152 2. Introduction to Machine Learning by Dr. Balaraman Ravindran, IIT Madras https://nptel.ac.in/courses/106106139 3. Machine Learning with Python by IBM https://cognitiveclass.ai/courses/machine-learning-with-python 4. Deep Learning by Prof. Prabir Kumar Biswas, IIT Kharagpur https://onlinecourses.nptel.ac.in/noc20_cs62/preview 5. Optimisation for Machine Learning : Theory and Implementation, IIT Delhi https://nptel.ac.in/courses/106106245
Relevant Topics for Self-study:
Paper References:
1. Gou, Jianping, Wenmo Qiu, Zhang Yi, Yong Xu, Qirong Mao, and Yongzhao Zhan. "A local mean representation-based K-nearest neighbor classifier." <i>ACM Transactions on Intelligent Systems and Technology (TIST)</i> 10, no. 3 (2019): 1-25. 2. Mudumba, Bharadwaj, and Md Faisal Kabir. "Mine-first association rule mining: An integration of independent frequent patterns in distributed environments." <i>Decision Analytics Journal</i> (2024): 100434. 3. Taye, Mohammad Mustafa. 2023. "Theoretical Understanding of Convolutional Neural Network: Concepts, Architectures, Applications, Future Directions" <i>Computation</i> 11, no. 3: 52. https://doi.org/10.3390/computation11030052.

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27)			
Data Science (DS)			
2671PC13: Advanced Data Management and Warehousing			
Semester	Credits	Teaching Scheme	Examination Scheme
1	4	L: 4 Hrs./ Week	CIE: 30 Marks ISE: 20 Marks ESE: 50 Marks
Prerequisite: Students should have prior knowledge of Discrete Mathematics, Database Management System.			
Course Objectives: The objective of this course is to provide students with - Understanding of database fundamentals. - Study of SQL and NoSQL databases. - Understanding of data warehouse architecture and enterprise-wide applications. - Design and preparing data for data warehouse development phases. - Understanding advances in Data warehouses.			
Course Outcomes: After completing this course, students will be able to: CO1: Design a database for real-world application. CO2: Apply suitable norms forms and formulate queries and PL/SQL for given relational database Application. CO3: Design a data warehouse model for real-world application. CO4: Illustrate recent developments in the Data warehouse.			
COURSE CONTENTS			
Module-I	The Database Environment and development process		14 Hrs.
Basic concepts: Data, Database System Structure, Types of databases like RDBMS, OORDBMS, NoSQL: document-oriented, graph. The Database approach: Data models, The components of database environment. The database development process: System development life cycle. Evolution of database system. Database application: Personal, multitier and Enterprise applications,Enhanced ER model,Universal data model. Case study: Design and Develop data modeling for any suitable application.			
Module-II	Database design		13 Hrs.

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

Relational Database design, Normalization during logical database design and reverse engineering Advanced Normal Forms. Implementation: Basic and Advanced SQL Parallel Databases: I/O Parallelism, Inter and Intra Query Parallelism, Inter and Intra Operation Parallelism. Distributed DBMS: Location transparency, synchronous and asynchronous, local and global transaction, commit protocol.		
Module-III	Big Data Descriptive Analytics	13 Hrs.
Master Data Management, Need for Data warehousing: Escalating need for strategic information, failure of past systems, operational versus decision support systems. Data Integration for Data warehousing :ETL process, Data Transformation. Descriptive analytics using data warehouse, Online analytical processing tools, SQL OLAP queries, MOLAP and ROLAP. Predictive and prescriptive analytics. Case Study- Design a data warehouse for a suitable system using an open-source tool.		
Module-IV	Advanced topics in Data Warehouse	12 Hrs.
Spatial and temporal Data Warehouse: General concepts of spatial databases, Conceptual modeling of spatial data warehouse, Manipulating temporal information in SQL, conceptual design of temporal data warehouse Graph data warehouse: Graph Data Models, Property Graph database system. Case Study: Industry relevance/recent trends		
Text Books:		
T1. Abraham Silberschatz, Henry F. Korth, and S. Sudarshan, <i>Database System Concepts</i>, McGraw-Hill Education, 2010, 6th Edition, ISBN:9332901384.		
T2. Jeff Hoffer, Ramesh Venkatraman, Heikki Topi, <i>Modern Database Management</i>, Twelfth Edition, Pearson, 2016, ISBN: 10-1-292-10185-7.		
T3. Jiawei Han & Micheline Kamber, <i>Data Mining–Concepts and Techniques</i>, Morgan Kaufmann Publishers, Elsevier, 2006, 2nd Edition ISBN: 9780123814791.		
T4. Paulraj Ponnaiah, <i>Data Warehousing Fundamentals</i>, Wiley, 2001, 2nd Edition, ISBN: 8126537299.		
Reference Books:		
R1. Connolly, T. M., & Begg, C. E. (2005), <i>Database systems: a practical approach to design, implementation, and management</i>, Pearson Education, 2005, 4th Edition, ISBN: 9788131720257.		
R2. Inmon, William, <i>Building the data warehouse</i>, H.. John wiley& sons, 2005, 3rd Edition, ISBN: 0-471-08130-2.		
R3. Vaisman, A., & Zimányi, E. (2014). <i>Data Warehouse Systems: Design and Implementation</i>, Springer, ISBN: 978-3-642-54655-6.		
Relevant MOOCs Course (Course name and Weblink)		

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

1. NPTEL 12 weeks course on "Database Management System" by By Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay | IIT Kharagpur Name: Dr. Parth Das
https://onlinecourses.nptel.ac.in/noc19_cs46/
2. Coursera 4 weeks course on " Data Warehouse Concepts, Design, and Data Integration" by University of Colorado System Instructor Name : Michael Mannino
<https://www.coursera.org/learn/dwdesign>
3. Coursera 4 weeks course on " Relational Database Support for Data Warehouses," offered by University of Colorado System Instructor Name : Michael Mannino
<https://www.coursera.org/learn/dwrelational>
4. NPTEL course on "Data Mining" offered by IIT Kharagpur Instructor Name: Prof.PabitraMitra
<https://nptel.ac.in/courses/106105174/>

Relevant Topics for Self-study:

Other Resources/Links:

1. <https://nifi.apache.org/>
2. <https://aws.amazon.com/redshift/>
3. <https://www.sas.com/>
4. <https://www.talend.com/>

Top 

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27)			
Data Science (DS)			
2661RM11: Research Methodology			
Semester	Credits	Teaching Scheme	Examination Scheme
1	3	L: 3 Hrs./ Week	CIE: 30 Marks ISE: 20 Marks ESE: 50 Marks
Prerequisite: Students should have prior knowledge of Mathematical foundations			
Course Objectives: The objective of this course is to provide students with - Understanding the philosophy of research in general. - Understanding diverse aspects of research methodologies. - Proficiency in both verbal and written presentation abilities. - Familiarity of Intellectual Property Rights (IPR) and plagiarism.			
Course Outcomes: After completing this course, students will be able to: CO1: Develop the significance of research and methodologies to collect and analyze data. CO2: Formulate hypotheses through experimentation. CO3: Design a comprehensive research report. CO4: Distinguish various forms of intellectual property rights.			
COURSE CONTENTS			
Module-I	Introduction to RM, defining the research problem and research design		14 Hrs.
Why do research? Meaning, objectives, Research types: computer science and engineering applications specific to nature of research in system building, research process, Criteria of good research. Formulating a research problem: Selecting a research problem, Writing research Objectives. Conceptualizing a research design: Meaning, Need and Basic principles of research design.			
Module-II	Literature survey, Data Collection and Sampling methods		09 Hrs.
Literature Survey: Importance, planning, locating relevant literature, managing information, and writing a review. Methods for primary and secondary data collection, data processing operations, measures for analyzing data, basics of sampling theory, concept of standard error, estimations on populations, determining sample size. Hypothesis testing: Parametric and non parametric methods.			
Module-III	Technical Content Writing		09 Hrs.

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43**M. Tech. (Data Science) Curriculum Structure****With effect from the A.Y. 2026-27**

Report writing significance and steps. Types of technical content publishing: White paper, journal, conference, poster, short paper etc. Writing various technical papers: survey paper, journal paper, indexing agencies, COPE.		
Module-IV	Intellectual Property Rights	12 Hrs.
Intellectual property: types of IP, IPR in India and abroad. Steps in patenting: Searching prior art, preparing the patent application, claims, filing, prosecution, objections, appeal, issuance or rejection of patent. Plagiarism and Research ethics. Research Tools: Plagiarism checking tools, Grammar checker tools.		
Text Books:		
T1. Ranjit Kumar, <i>Research Methodology- a step-by-step guide for beginners</i> , SAGE Publication, 2014, 4th Edition, ISBN: 9789351501336.		
T2. C. R. Kothari, <i>Research Methodology- Methods and Techniques</i> , New Age International Publishers, 2019, 4th Edition, ISBN: 9789386649225.		
Reference Books:		
R1. David V. Thiel, <i>Research Methods for Engineers</i> , Cambridge University Press., 2017, 2nd Edition, ISBN: 9781139542326.		
R2. C.G. Thomas, <i>Research methodology and scientific writing</i> , Springer, 2021, 2nd edition, ISBN: 978-3-030-64864-0.		
R3. <i>Professional Programme Intellectual Property Rights</i> , Law and Practice, The Institute of Company Secretaries of India, Statutory Body Under an Act of Parliament, September 2013: https://www.icsi.edu/media/webmodules/IPRLP_NOV29.pdf		
Paper References:		
1. Bayoudh, K., Knani, R., Hamdaoui, F. <i>et al.</i> A survey on deep multimodal learning for computer vision: advances, trends, applications, and datasets. <i>The Visual Computer</i> , 38(8), 2939–2970 (2022). https://doi.org/10.1007/s00371-021-02166-7 2. Taherdoost, H. (2016). Sampling methods in research methodology; how to choose a sampling technique for research. <i>How to Choose a Sampling Technique for Research</i> (April 10, 2016). 3. Romer, Paul. When should we use intellectual property rights?. <i>American Economic Review</i> 92.2 (2002): 213-216. 4. Wang, Yuntao, et al. A survey on ChatGPT: AI-generated contents, challenges, and solutions. <i>IEEE Open Journal of the Computer Society</i> (2023).		
Relevant MOOCs Course (Course name and Weblink)		
1. Research Methodology :Prof. Edamana Prasad, Prof. Prathap Haridoss, IIT Madras https://onlinecourses.nptel.ac.in/noc24_ge21/preview Research Methodology :Prof. Soumitro Banerjee, IISER Kolkata https://onlinecourses.nptel.ac.in/noc22_ge08/preview 2. Introduction to Statistical Hypothesis Testing: Dr. Arun Tangirala IIT Madras https://nptel.ac.in/courses/103106120 3. Roadmap for patent creation: Dr. Gouri Gargate IIT Kharagpur https://nptel.ac.in/courses/127105008		

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

Relevant Topics for Self-study:
Other Resources/Links:
1. The world intellectual Property Organization: https://www.wipo.int/portal/en/index.html 2. IP India: http://www.ipindia.nic.in/ 3. Cell For IPR Promotion and Management : http://cipam.gov.in/ 4. Draft patent rules: http://cipam.gov.in/wp-content/uploads/2018/12/Draft-Patent-Rules-2018.pdf 5. Manual of Patent Office Practice and Procedure: http://www.ipindia.nic.in/writereaddata/Portal/Images/pdf/Manual for Patent Office Practice and Procedure .pdf 6. WIPO IPR Resources: https://www.wipo.int/reference/en/

Top 

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27)			
Data Science (DS)			
2671PE13: Natural Language Processing			
Semester	Credits	Teaching Scheme	Examination Scheme
1	3	L: 3 Hrs./ Week	CIE: 30 Marks ISE: 20 Marks ESE: 50 Marks
Prerequisite: Students should have prior knowledge of Theory of Computation, Elementary Probability and Statistics, Basic understanding of Machine Learning			
Course Objectives: The objective of this course is to provide students with - Understand fundamental concepts of Computational Linguistics. - Understand various text pre-processing methods like tagging, parsing and semantic analysis. - Study neural network models for processing of natural language. - Get acquainted with modern approaches and applications in Computational Linguistics.			
Course Outcomes: After completing this course, students will be able to: CO1: Apply basic text processing techniques for given input. CO2: Analyze syntax and semantics of natural language data. CO3: Design neural language models for NLP application. CO4: Design machine translation and dialogue system.			
COURSE CONTENTS			
Module-I	Introduction		12 Hrs.
Introduction: Computing, Natural Language Processing and Linguistics, Stages of NLP Text Pre-processing: sentence segmentation, tokenization, feature extraction, issues in tokenization for different languages, word segmentation, text segmentation, normalization, case folding, Morphology, Stemming: Porters Algorithm, lemmatization. Spelling correction - dynamic programming approach for finding edit distance, N-gram Language Modeling- context sensitive spelling correction probabilistic language model, auto completion prediction.			
Module-II	Tagging, Syntax and Semantics		09 Hrs.
Sequence Labeling for Parts of Speech, HMM for speech tagging, Viterbi Algorithm, Conditional Random Fields. Syntax: Constituency and dependency parsing, Constituency parser -Syntactic structure, Parsing methodology, Different parsing algorithms, Parsing in case of ambiguity, Probabilistic parsing, CKY algorithm, Issues in parsing, Dependency parsing- Syntactic structure, Parsing methodology,			

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43**M. Tech. (Data Science) Curriculum Structure****With effect from the A.Y. 2026-27**

Transition-Based Dependency Parsing, Graph-Based dependency parsing, Evaluation, Co-reference resolution		
Semantics: Word Senses, Word relations, Word similarity and thesaurus methods, Named Entity Recognition, Word sense disambiguation, WordNet. Lexical and Distributional Semantics - Introduction, models of semantics, applications, Semantic Parsing - Problems and Techniques		
Pragmatics: Structures - Figures of Speech , Implicature (Speech Acts) , Sarcasm , Use of Common Sense Knowledge , Understanding Fiction , Domain specific meaning		
Module-III	Neural Language Models, Large Language Models	09 Hrs.
Overview of Machine Learning: Basics of Neural Networks, Feedforward Neural Network for NLP classification task, word embeddings, Recurrent Neural Networks (RNNs) as language model, Long short-term memory (LSTMs) network		
Transformers-Self attention network, Large Language Models: with transformers, Generation by sampling, training transformer.		
Fine-tuning and Masked language models: Bidirectional transformer encoders, training bidirectional encoders, fine tuning language models		
Module-IV	Applications - Machine Translation and Dialogue System	09 Hrs.
Language Divergences and Typology, Machine Translation using encoder decoder, Back-Translation, MT Evaluation		
Chatbots and Dialogue System: Properties of Human Conversation, Frame-Based Dialogue system, Chatbot, Dialogue system design		
Transfer learning via Prompting, Chain of Thought Prompting, Tree of Thought Prompting and Instruction Tuning, Retrieval Augmented Generation (RAG) techniques, Using vector datastore for RAG.		
Case Studies of industry relevance/recent trends - e.g. Summarization , Style transfer etc.		
Text Books:		
T1. Daniel Jurafsky and James H. Martin, <i>Speech and Language Processing</i> , Prentice Hall, 2008, 2nd Edition, ISBN: 978-0131873216.		
T2. Allen James, <i>Natural Language Understanding</i> , Benjamin/Cumming, 1994, 2nd Edition, ISBN: 978-0805303346.		
T3. Chris Manning and Hinrich Schuetze, <i>Foundations of Statistical Natural Language Processing</i> , MIT Press, 1999, 1st Edition, ISBN: 978-0262133609.		
T4. Jacob Perkins, <i>Python 3 text processing with NLTK 3 cookbook</i> , Packt Publishing Ltd., 2014, 2nd Edition, ISBN-13: 9781782167853.		
Reference Books:		
R1. Jacob Eisenstein, <i>Natural Language Processing</i> , The MIT Press, 2019, 1st Edition, ISBN: 9780262042840.		
Paper References:		
1. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin, "Attention is all you need," in Proceedings of the 31st		

International Conference on Neural Information Processing Systems (NIPS'17),pp. 6000–6010,2017,doi:10.5555/3295222.3295349 - Pradhan, R., Sharma, D.K., "An ensemble deep learning classifier for sentiment analysis on code-mix Hindi-English data," in <i>Soft Comput</i> 27, 11053 (2023),doi:10.1007/s00500-022-07091-y - Jessica López Espejel, El Hassane Ettifouri, Mahaman Sanoussi Yahaya Alassan, El Mehdi Chouham, Walid Dahhane, "GPT-3.5, GPT-4, or BARD? Evaluating LLMs reasoning ability in zero-shot setting and performance boosting through prompts," in <i>Natural Language Processing Journal</i>,Volume 5,2023,doi:10.1016/j.nlp.2023.100032 - Lee, Jooyoung, Fan Yang, Thanh Tran, Qian Hu, Emre Barut, Kai-Wei Chang and Chengwei Su., "Can Small Language Models Help Large Language Models Reason Better?: LM-Guided Chain-of-Thought," <i>ArXiv</i> abs/2404.03414 (2024) - Ramchandra Joshi, Rusbabh Karnavat, Kaustubh Jirapure, and Raviraj Joshi. 2020. Domain Adaptation of NMT models for English-Hindi Machine Translation Task : AdapMT Shared Task ICON 2020. In <i>Proceedings of the 17th International Conference on Natural Language Processing (ICON)</i>, pages 11–16, Patna, India. NLP Association of India (NLP AI). R. Joshi, "L3cube-mahacorporus and mahabert: Marathi monolingual corpus, marathi bert language models, and resources," in <i>Proceedings of the WILDRE-6 Workshop within the 13th Language Resources and Evaluation Conference</i>, 2022, pp. 97–101 - Facundo Carrillo, Guillermo A. Cecchi, Mariano Sigman, Diego Fernández Slezak, "Fast Distributed Dynamics of Semantic Networks via Social Media,," <i>Computational Intelligence and Neuroscience</i>, vol. 2015,2015,doi: 10.1155/2015/712835.
Relevant MOOCs Course (Course name and Weblink)
- Course Title: Natural Language Processing Specialization offered by Kaiser, Coursera. Link: https://www.coursera.org/specializations/natural-language-processing - Course Title: Applied Natural Language Processing offered by Swayam, NPTEL Link: https://onlinecourses.nptel.ac.in/noc20_cs87/preview
Relevant Topics for Self-study:
-
Other Resources/Links:
https://www.ibm.com/topics/instruction-tuning https://www.ibm.com/products/watsonx-ai

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27) Data Science (DS) 2671VS31 : Laboratory Proficiency -I (LP-I)			
Semester	Credits	Teaching Scheme	Examination Scheme
1	2	P: 4 Hrs./ Week	CIE/TW: 25 Marks ESE (OR): 25 Marks
Prerequisite: Students should have prior knowledge of Knowledge of programming languages, Basics of Python/R/Java			
All assignments are compulsory. Each student should implement the assignments individually. Laboratory teachers should make sure that the dataset/code/write up is not the same.			
Course Objectives: The objective of this course is to provide students with - Explore graph theory in computational problem-solving. - Use descriptive statistical measures for conducting statistical inference. - Explore data design and skills for data warehouse development phases. - Examine NoSQL databases for their functionality and applications. - Develop IoT applications utilizing embedded devices. - Engineer security solutions tailored for real-world applications. - Explore different approaches to Shared Memory and Inter-process Communication. - Comprehend the fundamental principles behind designing Distributed File Management systems.			
Course Outcomes: After completing this course, students will be able to C01: Apply Graph Theory for problem modeling and resolution. C02: Analyze population characteristics through central tendency, dispersion, and relationships. C03: Analyze the performance of different prediction algorithms for given application. C04: Design schema and execute database queries using NoSQL. C05: Design & store data in a data warehouse using open-source tool. C06: Develop IoT applications leveraging embedded devices. C07: Apply cryptographic techniques and tools for problem-solving applications. C08: Apply various NLP techniques to given application. C09: Implement process and file management system calls using C/Rust.			
All assignments are compulsory. Each student should implement the assignment individually. Every student should use a different data set.			
COURSE CONTENTS			
Expt. No.	Problem Statement	CO's	
1.	Mathematical Foundations for Computing a. Write a program to represent user's information in social networks using graphs. Find the number of associations of each user. OR	C01	

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

	b. Design an algorithm to organize the sequence of courses within engineering programs, leveraging graph theory to account for the prerequisite dependencies among courses.	
2.	Write a program, in the language of your choice, to calculate measures of central tendency, dispersion and relationship of given data.	C02
3.	Advanced Machine Learning Download Email spam classification dataset from Kaggle https://www.kaggle.com/datasets/balaka18/email-spam-classification-dataset-csv. Email Spam detection has two states: a) Normal State – Not Spam, b) Abnormal State – Spam. Use K-Nearest Neighbors and Support Vector Machine to classify the email. Extract confusion matrix from the test results and compare the performance of both.	C03
4.	Download Housing Rent Prediction Dataset from Kaggle and predict the house rent using regression. https://www.kaggle.com/datasets/iamsouravbanerjee/house-rent-prediction-dataset/	C03
5.	Advanced Data Management and Warehousing For an organization of your choice, choose a set of business processes. - Design star/ snowflake schemas for analyzing these processes. Create a fact constellation schema by combining them. Eg. University database, Manufacturing Industry. - Use an open source ETL tool to extract data from different data sources, apply suitable transformations and load into destination tables. Eg. Manufacturing Industry- data of sales, products, purchase.	C04 C05
6.	Implement the following NoSQL queries for the product review database. - Create a collection product review database. - Insert 5 documents in the collection. - Find the product list sorted by price of the product from lowest to highest. Skip the first two documents. - Delete product review document from the collection where the rating is less than two and product type is 'camera'.	C04 C05
7.	IoT Architecture & Protocols Design and develop a home automation security system using embedded devices.	C06
8.	Design and develop a remote health monitoring system using IoT devices and ThingSpeak cloud.	C06
9.	Information security Implement the Diffie-Hellman Key Exchange mechanism using HTML & JavaScript. Consider the end user as one of the parties (Alice) & the JavaScript application as another party (bob). Also demonstrate MITM (man in the middle attack) on it.	C07

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

10.	a. Write a program for Elliptic curve cryptography. OR b. Write a program to apply Advanced Encryption Standard (AES) Algorithm for a practical application like URL Encryption.	C07
11.	Natural Language Processing Implement complete natural language processing pipeline for Indic language text input.	C08
12.	Implement Conversational interface (chatbot) using python.	C08
13.	Distributed Operating System Develop a program in C or Rust to implement Process Management System Calls, which enable the creation, manipulation, and control of processes within an operating system environment.	C09
14.	Design and implement a Distributed File System (DFS) like Hadoop Distributed File System (HDFS), capable of storing and managing large volumes of data across multiple nodes in a distributed environment.	C09
15.	Mini Project (Based on Core or Elective subjects).	

[Top](#) 

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27)			
Data Science (DS)			
2661RM31: Research and Technical Communication Laboratory(RTCL)			
Semester	Credits	Teaching Scheme	Examination Scheme
1	1	P: 2 Hrs./ Week	CIE / TW : 25 Marks
Prerequisite: Students should have prior knowledge of Basic statistics			
Course Objectives: The objective of this course is to provide students with - Understanding of research papers. - Formulation of research questions, hypotheses and derive research objectives. - Enhancing proficiency in both verbal and written presentation abilities. - Getting familiarity with research proposals and patent writing.			
Course Outcomes: After completing this course, students will be able to			
CO1: Review research papers based on motivation, methodology, analysis and conclusion.			
CO2: Analyze research data using relevant statistical techniques.			
CO3: Interpret the research findings.			
CO4: Communicate research findings through written reports and oral presentations.			
COURSE CONTENTS			
Expt. No.	Problem Statement		CO's
Ethical practices and Websites to be used in following assignments.			
1.	Reviewing Research papers a. Write a set of research questions for the case study of your interested topic. Construct the set of research objectives from the set of questions. b. Utilize academic search tools to extract pertinent journal papers. Engage in comprehensive reading of the paper, focusing on the following aspects: major contribution, datasets utilized, methodology employed, analysis and outcomes, conclusions drawn, and any limitations identified. Post-review, articulate the research question believed to have been addressed by the author. Evaluate whether the paper effectively substantiates its conclusions in addressing the aforementioned question.		CO1
2.	Literature Review Quality and research proposal a) Select a journal paper within your engineering discipline and provide a detailed summary of its literature review section, comprising 400 to 500 words. Evaluate the quality and relevance of the papers cited within the review, assessing the commentary		CO4

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

	on their contributions to the broader field. Additionally, analyze any notable omissions of papers that hold significant importance within the field. b) Craft a new research proposal based on insights gleaned from a chosen published journal paper. Focus specifically on scrutinizing the discussion and conclusion sections of the paper to identify suggestions for future research endeavors.	
3.	Data modeling and statistical review a) Download a set of weather data from the Internet covering the temperature and atmospheric pressure over five days period. Present the data using 2D and 3D plots, and so deduce if the weather conditions are trending either higher or lower over these five days period. b) Numerical modeling: Find a research paper that employs numerical modeling to validate experimental findings. Compare and contrast the differences observed between the experimental and modeling results. Evaluate whether the authors have provided insights into the accuracy of both the experimental and modeling methodologies. Provide recommendations for enhancing the quality of the modeling techniques presented in the paper. c) Statistical review: Within your engineering domain, examine a published paper incorporating statistical analysis. Draft a concise report delineating the statistical methods employed. Propose enhancements to the statistical analysis. Recommend additional parameters that could have been captured during data acquisition and elucidate how to analyze the comprehensive dataset to ascertain the influence and statistical significance of these supplementary measurements.	C02
4.	Research Paper analysis 1. Note the keywords and type them into one of the web-based academic search engines (e.g. googlescholar.com). a. Does the original article appear in the search results? b. Compare the citations of this paper with those from the most highly cited paper in the search results. c. If this paper was published before your original article, is it cited in your article? Do you think this high-cited paper should have been listed as a reference in your original article? Give reasons for your decision. 2. How many citations does this article have? 3. Have the same authors published further work in this field? a. How many citations does this highly cited article have?	C03
5.	Mini Project: I. Generate a novel research proposal based on an existing published journal paper by following these steps: 1. Begin with an introduction that outlines the research problem and objectives.	C04

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

	2. Provide background information and rationale to contextualize the proposed research. 3. Detail the methodology, including research design, data collection methods, and analytical techniques. 4. Write a research plan and budget requirements. 5. Summarize the conclusions drawn from the existing paper and propose areas for future research. 6. Compile a bibliography listing all the references cited in the proposal. II. Analyze the paper's key findings and identify significant outcomes. Structure these outcomes in patent claims, ensuring they represent novel advancements not readily deduced from prior research.	
--	---	--

[Top](#) 

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43
M. Tech. (Data Science) Curriculum Structure
With effect from the A.Y. 2026-27

First Year M. Tech (F. Y M. Tech) AY (2026-27)			
Data Science (DS)			
2671VS32: Programming Proficiency Laboratory			
Semester	Credits	Teaching Scheme	Examination Scheme
1	1	P: 2 Hrs./ Week	CIE / TW: 25 Marks
Prerequisite: Students should have prior knowledge of Basics of Programming			
Course Objectives: The objective of this course is to provide students with - Knowledge of Python/R/Java programming. - Applications such as object oriented Files, Web Scraping, Data Preprocessing, Data Visualization, Natural Language Processing, Machine Learning, Graphical User Interface. - Explore functions in Python/R/Java for evaluating performance of the designed model.			
Course Outcomes: After completing this course, students will be able to CO1: Write a program in Python/R/Java language. CO2: Implement File Handling, Web Scraping using Python/R/Java language. CO3: Write a program for Data Preprocessing, Data Visualization using Python/R language. CO4: Develop HCI Interfaces in Python/R/Java language. CO5: Analyze performance of the designed model by using functions in Python/R/Java.			
COURSE CONTENTS			
Expt. No.	Problem Statement	CO's	
1.	A. Design a program that retrieves and lists the names of all regional languages along with the number of related articles in the order they are presented on Wikipedia.org B. Develop a tool to extract and organize images from a specific web page URL, providing options for filtering, sorting, and downloading the images based on various criteria such as resolution, file type, or metadata.	CO1, CO2	
2.	A. Write a program to perform various data preprocessing operations such as handling missing values, one hot encoding, label encoding & data normalization on the following dataset. Dataset: https://rb.gy/ppmnxv B. Write a program to reduce the number of features in the following dataset while preserving most of the important information. Use techniques such as Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), or feature selection methods like Recursive Feature Elimination (RFE) for this purpose. Dataset: https://shorturl.at/aejsM .	CO1, CO3	

SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE-43

M. Tech. (Data Science) Curriculum Structure

With effect from the A.Y. 2026-27

3.	A. Write a program to visualize educational data such as student performance, attendance, and learning outcomes to identify areas for improvement, track progress over time, and personalize learning experiences. B. Design a data visualization dashboard to analyze the monthly sales performance of a retail store over the past year. The dashboard should include visualizations of total sales, average sales per day, and sales trends over time. Additionally, it should identify the best and worst performing months and provide insights into factors affecting sales fluctuations.	C01, C04
4.	A. Write a program to explore different tokenization techniques such as whitespace tokenization, word tokenization, or sentence tokenization. B. Write a program to implement the Porter Stemmer algorithm for Stemming.	C01,
5.	A. Write a program to predict the house price using a regression algorithm in machine learning. Evaluate the performance of the model using various performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Huber Loss & R2 score. Link to dataset: https://rb.gy/ppmrxv B. Write a program to split the Iris flower dataset into 80% train data and 20% test data. Fit the data into the Naive Bayes model for training and testing. Plot the Confusion Matrix. Evaluate the performance using various performance metrics such as Accuracy, Precision, Recall & F1 Score.	C01, C05
6.	A. Write a mobile application to design a BMI (Body Mass Index) calculator application using tkinter library in Python / shiny library in R where users can input their height and weight, and the application calculates and displays their BMI along with a corresponding interpretation (e.g., underweight, normal weight, overweight). B. Build an application using the tkinter library in Python / shiny library in R that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The application should have embedded objects, widgets etc.	C01, C05

Text Books:

T1. Reema Thareja, *Python Programming Using Problem Solving Approach*, Oxford University Press, 2017, 1st Edition, ISBN 13: 978-0-19-948017-6.

T2. R. Nageswara Rao, *Core Python Programming*, Dreamtech Press, 2018, 2nd Edition, ISBN-13: 978-9386052308.

T3. Andrie de Vries, *R For Dummies*, John Wiley & Sons (Wiley), 2015, 2nd edition, ISBN-101119055806.

Reference Books:

R1. Martin C. Brown, *Python: The Complete Reference*, McGraw Hill Education, 2018, 1st Edition, ISBN-13: 978-9387572942.

R2. Romano Fabrizio, *Learning Python*, Packt Publishing Limited, 2015, 1st Edition, ISBN:

9781783551712, 1783551712.

R3. Paul Barry, *Head First Python- A Brain Friendly Guide*, SPD O'Reilly, 2016, 2nd Edition, ISBN:978-93-5213-482-3.

Relevant MOOCs Courses:

1. SWAYAM - Programming in Python by Dr. Rizwan Rehman, Dibrugarh University.
https://onlinecourses.swayam2.ac.in/cec22_cs20/preview
2. NPTEL - The Joy of Computing using Python by Prof. Sudarshan Iyengar, IIT Ropar
https://onlinecourses.nptel.ac.in/noc24_cs57/preview
3. IBM - Data Visualization using python.
<https://cognitiveclass.ai/courses/data-visualization-python>
4. NPTEL - Introduction to R Software by Prof. Shalabh , IIT Kanpur
https://onlinecourses.nptel.ac.in/noc19_ma33/preview

[Top](#) 